

SQL & MySQL Assignment: Lab-1

1. What is SQL and Its Use?

SQL, or Structured Query Language, is the standard language used to communicate with relational database systems. It allows users to store, manipulate, and retrieve data efficiently. The primary purpose of SQL is to provide a consistent way to manage large sets of structured information.

Its uses include:

- Data Retrieval: Extracting specific information from large datasets using queries.
- Data Manipulation: Adding new records, updating existing ones, or deleting unnecessary data.
- Database Definition: Creating and modifying tables, schemas, and other structures.
- Access Control: Granting or revoking permissions to ensure data security.
- Transaction Management: Ensuring data consistency and reliability when multiple operations are performed.

2. Describe all the different commands of SQL, mention use of it and syntax.

SQL commands are broadly divided into categories based on their purpose:

- Data Definition Language (DDL): These commands define and modify the structure of the database.

CREATE is used to create new tables or databases. Example:

sql

```
CREATE TABLE Students (ID INT, Name VARCHAR(50));
```

- **ALTER** modifies the structure of an existing table, such as adding a new column.

sql

```
ALTER TABLE Students ADD Age INT;
```

- **DROP** deletes a table or database permanently.

sql

```
DROP TABLE Students;
```

- Data Manipulation Language (DML): These commands handle the actual data inside tables.

INSERT adds new records.

sql

```
INSERT INTO Students VALUES (1, 'Lakshay', 20);
```

- **UPDATE** modifies existing records.

sql

```
UPDATE Students SET Age = 21 WHERE ID = 1;
```

- **DELETE** removes records.

sql

```
DELETE FROM Students WHERE ID = 1;
```

- Data Query Language (DQL):

SELECT retrieves data from tables.

sql

```
SELECT Name, Age FROM Students WHERE Age > 18;
```

- Transaction Control Language (TCL): These commands manage transactions to ensure data integrity.
 - **COMMIT** saves changes permanently.
 - **ROLLBACK** undoes changes made in the current transaction.

- **SAVEPOINT** sets a checkpoint within a transaction.
- Data Control Language (DCL): These commands manage user privileges.
 - **GRANT** provides access rights.
 - **REVOKE** removes access rights.

3. Write step by step procedure to install MYSQL.

Step 1: Go to the official MySQL Community Downloads page and select the latest MySQL Installer version for Windows.

Step 2: Choose the installer type. You can either download the smaller web installer (which fetches components during setup) or the full installer (which contains all components).

[General Availability \(GA\) Releases](#)
[Archives](#)

MySQL Installer 8.0.35


Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:

Select Operating System:

Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.35.0.msi)	8.0.35	2.1M	Download
MD5: 214df2ccdf83eb5edc6ca7c115792406 Signature			
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.35.0.msi)	8.0.35	288.6M	Download
MD5: 2cfd448a2971b6b5323775ef9e8d012 Signature			

Step 3: When prompted, you may log in or sign up for an Oracle account. If you prefer, click on *"No thanks, just start my download"* to continue without an account

MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

Login »
using my Oracle Web account

Sign Up »
for an Oracle Web account

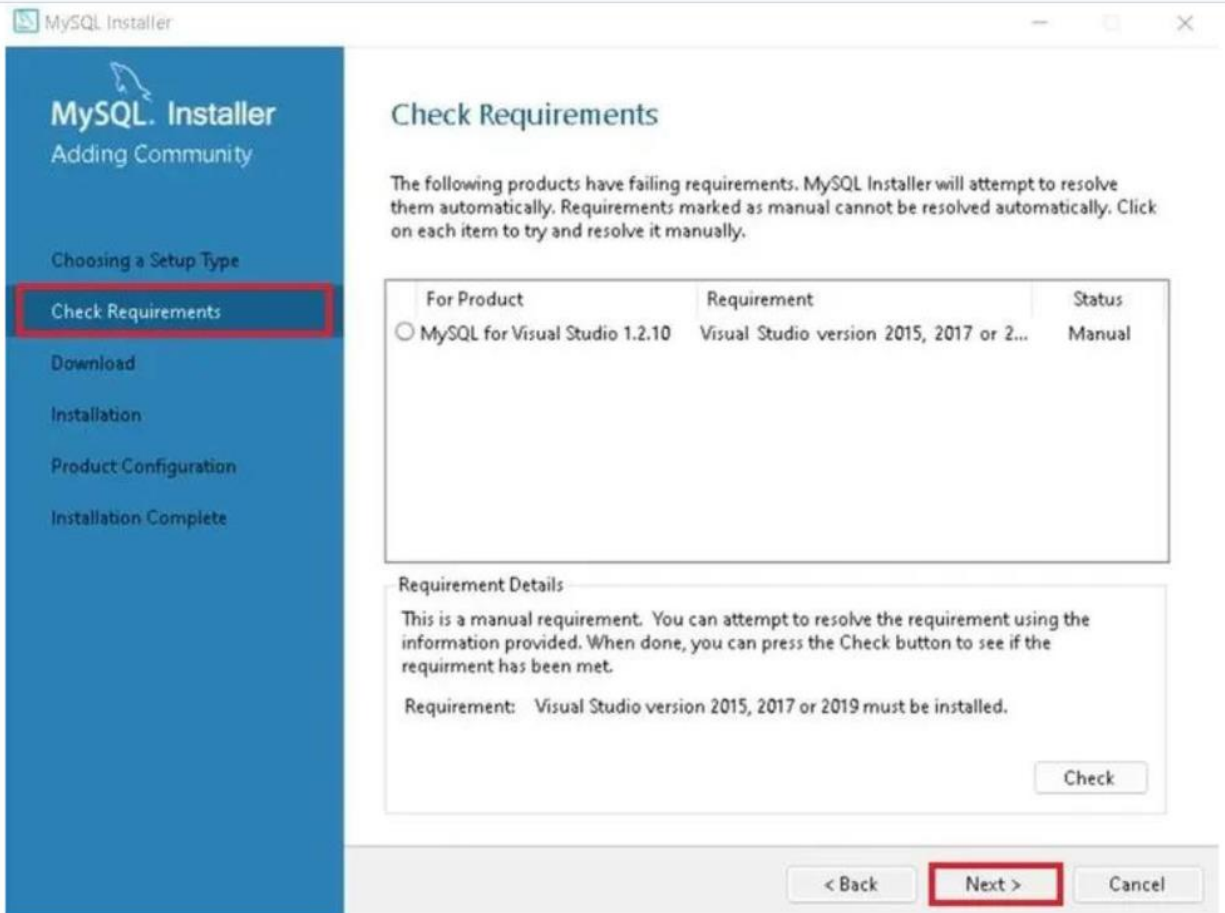
MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

No thanks, just start my download.

Step 4: Once downloaded, run the **.msi** installer file to launch the MySQL Installer wizard.

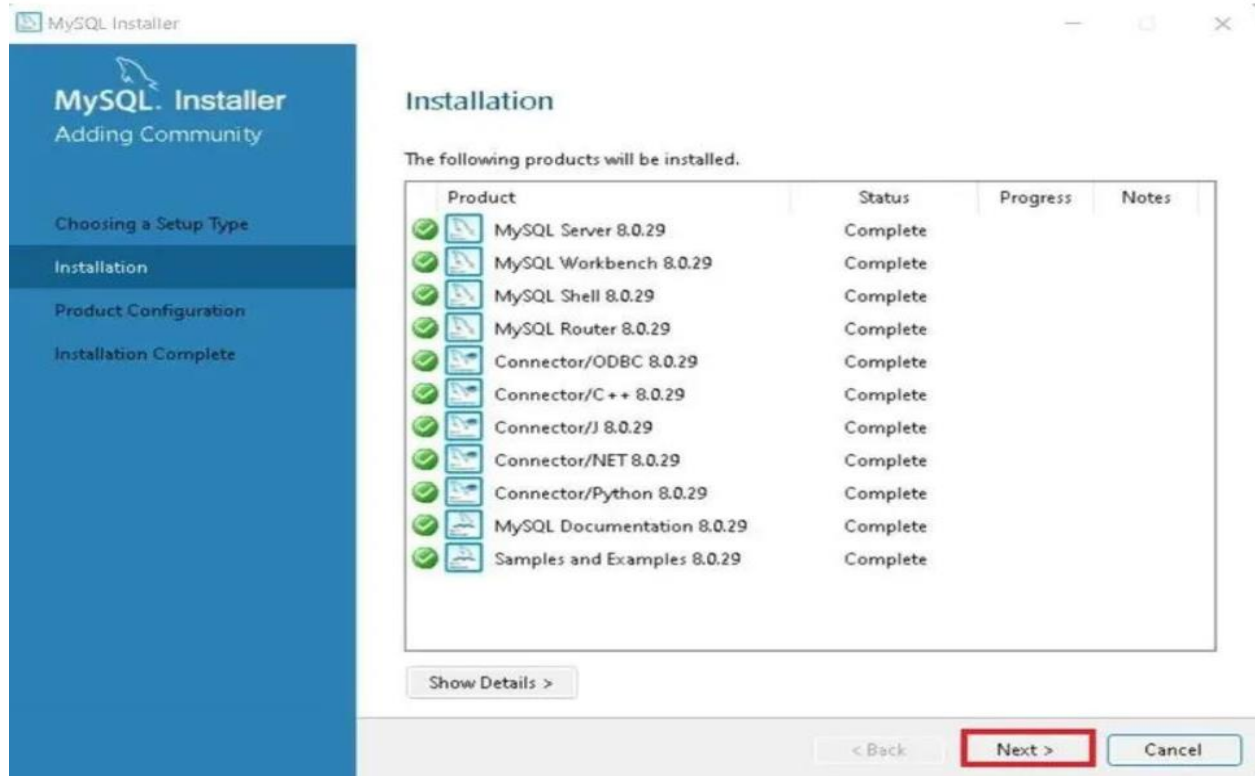
Step 5: Select the setup type. Options include Developer Default, Server Only, Client Only, or Custom. Choose the one that suits your requirement.

Step 6: The installer will check system requirements. If any prerequisites are missing (such as Visual Studio for MySQL for Visual Studio integration), resolve them before proceeding.

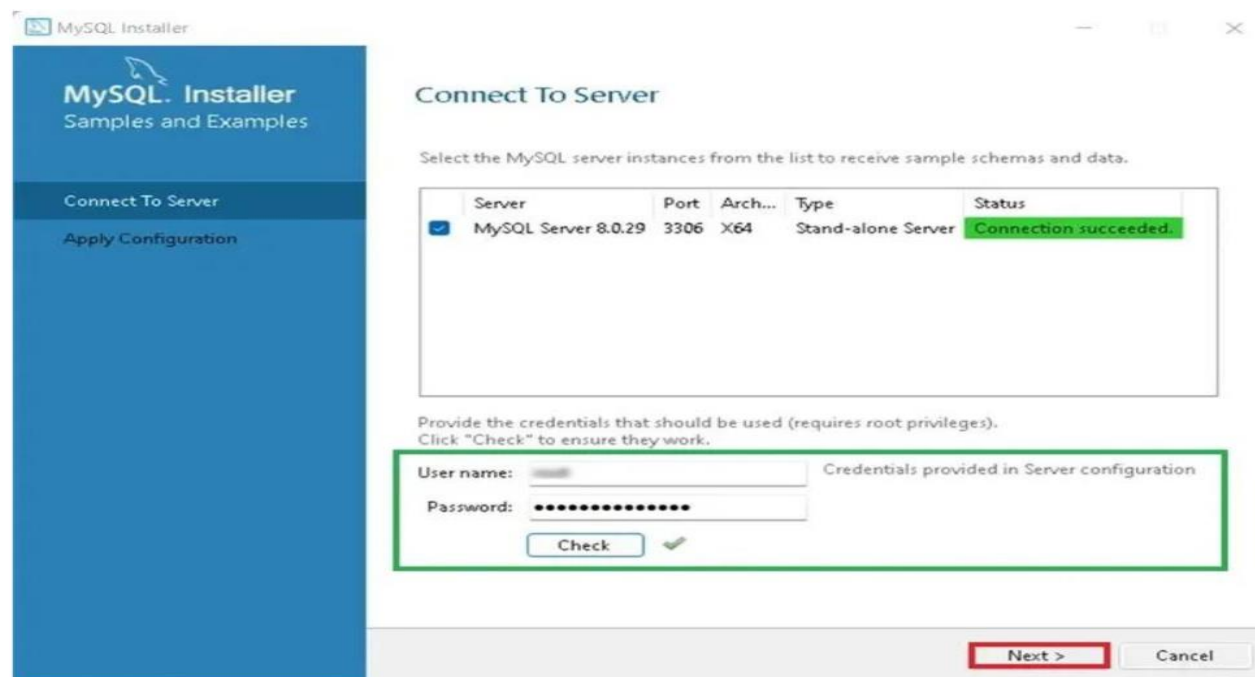


Step 7: The installer will now download and install the selected MySQL products such as MySQL Server, Workbench, Shell, Router, Connectors, Documentation, and Samples.

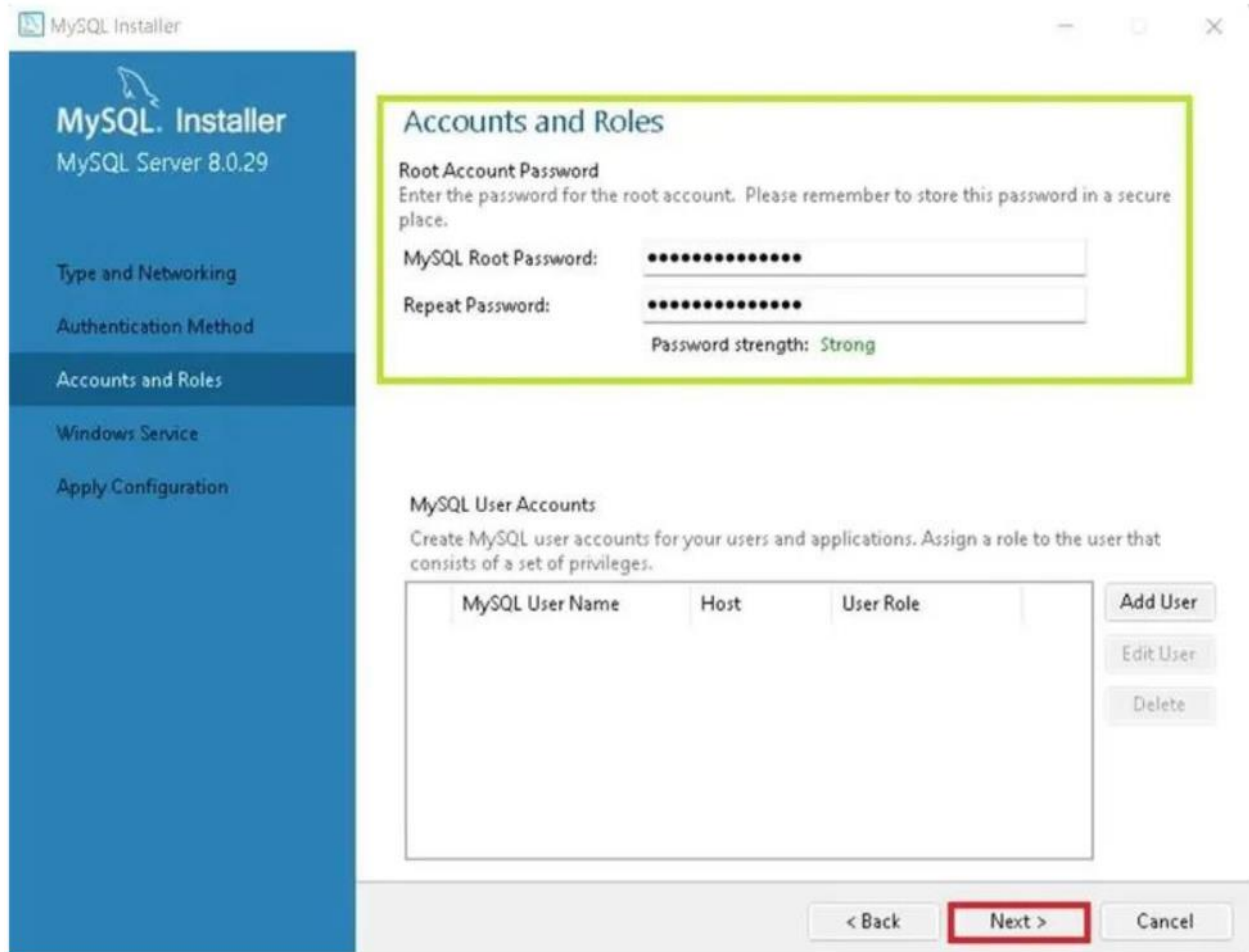
Step 8: Configure the MySQL Server type and networking. Select whether the server will run as a standalone or clustered server, and confirm the default port (3306).



Step 9: Choose the authentication method. Strong password encryption is recommended for better security.



Step 10: Set up the root account by creating a strong password. You may also add additional user accounts and assign roles to them.



The screenshot shows the 'MySQL Installer' window for 'MySQL Server 8.0.29'. The left sidebar lists the installation steps: 'Type and Networking', 'Authentication Method', 'Accounts and Roles' (which is the current step), 'Windows Service', and 'Apply Configuration'. The main area is titled 'Accounts and Roles' and contains two sections. The first section, 'Root Account Password', prompts the user to enter a password for the root account, with a note to store it securely. It includes input fields for 'MySQL Root Password' and 'Repeat Password', both masked with dots. Below these fields, it indicates 'Password strength: Strong'. The second section, 'MySQL User Accounts', instructs the user to create accounts and assign roles. It features a table with columns for 'MySQL User Name', 'Host', and 'User Role'. To the right of the table are buttons for 'Add User', 'Edit User', and 'Delete'. At the bottom of the window, there are navigation buttons: '< Back', 'Next >' (which is highlighted with a red box), and 'Cancel'.

MySQL Installer
MySQL Server 8.0.29

Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password strength: Strong

MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role
-----------------	------	-----------

Add User
Edit User
Delete

< Back **Next >** Cancel

Step 11: Configure MySQL as a Windows Service so that it starts automatically whenever the system boots.

Step 12: Apply the configuration. The installer will test the server connection, finalize the setup, and display a completion screen. You can then launch MySQL Workbench or MySQL Shell to begin using MySQL.

MySQL. Installer

Adding Community

Choosing a Setup Type

Installation

Product Configuration

Installation Complete

Installation Complete

The installation procedure has been completed.

Copy Log to Clipboard

- ☒ Start MySQL Workbench after setup
- ☒ Start MySQL Shell after setup

The MySQL Shell is an advanced MySQL client application that can be used to work with single MySQL Server instances. Further, it can be used to create and manage InnoDB Cluster, an integrated solution for high availability and scalability of MySQL databases, without requiring advanced MySQL expertise.



Refer to the following links for documentation, tutorials and examples on MySQL Shell:

[MySQL Shell Documentation](#)

[Setting up a Real World Cluster Blog](#)

[The All New MySQL InnoDB ReplicaSet Blog](#)

[Changing Cluster Options Live Blog](#)

Finish